Home Equity Loan Default Model

Interpretable logistic regression for borrower-level default risk

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5,960
LOANS ANALYZED

19.95%
OBSERVED BAD-LOAN RATE

0.7853
FIVE-VARIABLE CV AUC

+0.0490
AUC LIFT VS BENCHMARK

EXECUTIVE SUMMARY

A compact probability-of-default model uses five economically meaningful borrower attributes to rank home-equity loan risk. The expanded specification improves discrimination over a three-variable benchmark while remaining transparent and easy to interpret.

METHOD

1. Median imputation and standardization inside each fold.
2. Stratified five-fold cross-validation with out-of-fold PDs.
3. Validation through AUC, Brier score, classification metrics and calibration deciles.

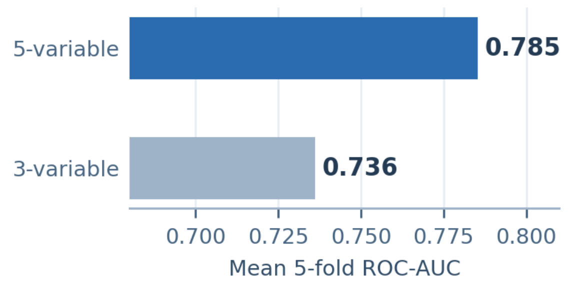
KEY FINDINGS

Discrimination: AUC rose from 0.7363 to 0.7853.
Strongest risk signal: DELINQ coefficient +0.784; odds ratio 2.19 per 1 SD.
Protective factor: CLAGE coefficient -0.550; odds ratio 0.58 per 1 SD.
Collinearity: Maximum VIF was 1.06.

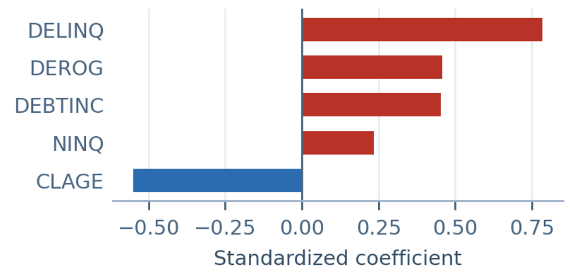
THRESHOLD IMPLICATION

At a 0.50 cutoff, specificity is 97.1% but sensitivity is only 30.4%. A production threshold should be chosen using approval economics, expected loss and risk appetite.

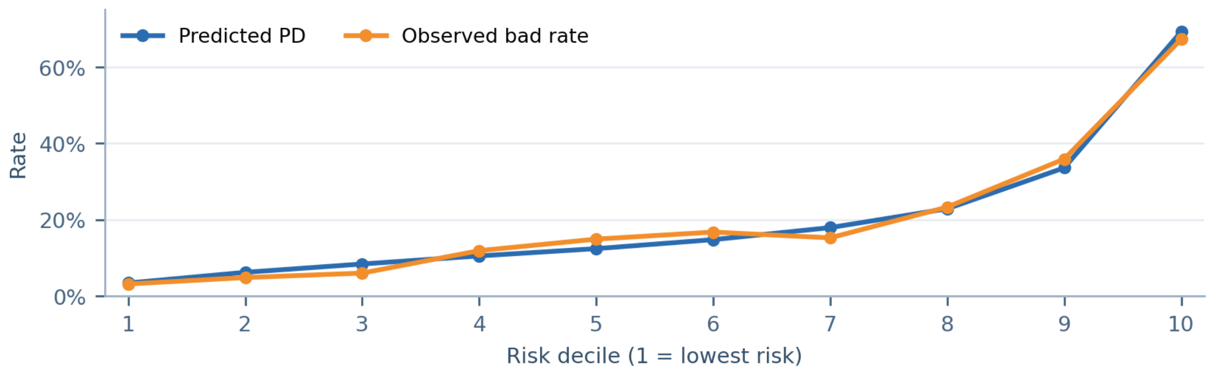
Model discrimination



Standardized risk drivers



Calibration by risk decile



Top decile: predicted PD 69.3% vs observed bad rate 67.3%; risk separation is clear across ranked groups.

Source: attached HMEQ dataset and reproducible Python model output. Educational portfolio project; not for lending decisions.

Logistic regression | 5-fold stratified CV | Out-of-fold validation